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Let  $C_{1,0}([0, 1]; \mathbb{R}^d)$  be the space of absolutely continuous  $d$ -dimensional paths  $X_{(\cdot)} : [0, 1] \rightarrow \mathbb{R}^d$  with the norm  $\|X\|_{1;[0,1]} := \int_0^1 |\dot{X}_s| ds$ . In this context, the reservoir computing phenomenon is usually understood as the phenomenon in which a sufficiently regular input/output system (time-invariant, causal, continuous)  $F : C_{1,0}([0, 1]; \mathbb{R}^d) \rightarrow C_{1,0}([0, 1]; \mathbb{R}^m)$  s.t.  $F(X)_t = F(X_{[0,t]})$  can be well approximated by the composition of: 1) a high dimensional linear state space system

$$dZ_t^X = \sum_{i=1}^d A'_i Z_t^X dX_t^i + B' dX_t, Z_t \in \mathbb{R}^N, A'_i \in \mathbb{R}^{N \times N}, B' \in \mathbb{R}^{N \times d}$$

whose dynamics are governed by random projections  $A_i, B$ , and 2) an optimal choice of a linear readout  $V \in \mathbb{R}^{m \times N}$ . That is, for a given  $\epsilon > 0$  the following holds:

$$\sup_{t \in [0,1]} \|F(X_{[0,t]}) - V Z_t^X\|_2 < \epsilon, \forall X_{(\cdot)} \in K \subset C_{1,0}([0, 1]; \mathbb{R}^d)$$

with  $K$  satisfying e.g. compactness.

There are many insightful results that shed light on why exactly this is possible. The following outline is an example of how we can arrive at this conclusion:

1. (Closure of solutions) First, the input-output systems of interest  $F(\cdot)_t \in C(K) = \{\}$ , are shown to be in the closure of solutions  $\{Z_t^{(\cdot)} \text{ with dynamics } A_i \in \mathbb{R}^{N \times N}, 1 \leq i \leq d, B \in \mathbb{R}^{N \times d}, N > 0\}$ . This is done by means of a classical theorem of universal approximation. In fact, the space  $C_{1,0}([0, t]; \mathbb{R})$  can be shown to be point-separable via the signature transform  $S(\cdot)_{[0,t]}^n : X \mapsto$

$(\int_{0 < t_{i_1} < \dots < t_{i_n} < t} dX_{t_{i_1}} \dots dX_{t_{i_n}})_{1 \leq i_1 \leq d, \dots, 1 \leq i_n \leq d} \in (\mathbb{R}^d)^{\otimes n}$ , the terms of which can be regrouped (in increasing order) to form an element of the tensor algebra  $T((\mathbb{R}^d)) = \prod_{n=0}^{\infty} (\mathbb{R}^d)^{\otimes n} \supset \bigoplus_{n=0}^{\infty} (\mathbb{R}^d)^{\otimes n} =: T(\mathbb{R}^d)$ , with  $T((\mathbb{R}^d))$  admitting a dual space  $T((\mathbb{R}^d))^*$  containing a subspace identifiable with  $T(\mathbb{R}^d)^* = \prod_{n=0}^{\infty} ((\mathbb{R}^d)^{\otimes n})^* \cong \prod_{n=0}^{\infty} ((\mathbb{R}^d)^*)^{\otimes n} \cong T((\mathbb{R}^d)^*)$ , the latter having an action defined by applying the functional in  $((\mathbb{R}^d)^{\otimes n})^*$  to the corresponding tensor in  $(\mathbb{R}^d)^{\otimes n}$  and summing across  $n = 0, 1, 2, \dots$ . The fact that for any two  $X, Y \in C_{1,0}([0, 1]; \mathbb{R}^d)$ , there exist  $f \in T(\mathbb{R}^d)^* \subset T((\mathbb{R}^d))^*$  such that

$$f((S(X)_{[0,t]}^0, S(X)_{[0,t]}^1, \dots)) \neq f((S(Y)_{[0,t]}^0, S(Y)_{[0,t]}^1, \dots))$$

(along with a few other arguments) allows us to apply the Stone-Weierstrass theorem to conclude that  $\{f' : X \subset K \mapsto f(X) | f \in T((\mathbb{R}^d))^*\} \subset C(K)$